OOP Deep Interview Questions

1. What is OOP?
Answer: OOP is a programming paradigm that organizes code into objects containing data and behavior.
2. Why use OOP?
Answer: To improve code reusability, maintainability, scalability, and modularity.
3. What is Encapsulation?
Answer: Encapsulation hides data and exposes it through controlled methods or properties.
4. Why is Encapsulation important?
Answer: It protects object state and prevents unauthorized access.
5. What is Abstraction?
Answer: Abstraction hides implementation details and exposes only essential functionality.
6. What is Inheritance?
Answer: Inheritance allows a child class to reuse and extend the functionality of a parent class.
7. What is Polymorphism?
Answer: Polymorphism allows the same method call to behave differently based on the object type.
8. Compile-time vs Runtime Polymorphism?
Answer: Overloading is compile-time; overriding is runtime.
9. Why prefer Composition over Inheritance?
Answer: Composition provides loose coupling and greater flexibility.
10. Can C# support multiple inheritance?
Answer: No, classes don't; interfaces do.

Interface vs Abstract Class

11. Interface vs Abstract Class?
Answer: Interface defines a contract, while an abstract class provides both common implementation and a contract.
12. When would you choose an Interface?
Answer: When unrelated classes need to implement the same behavior.
13. When would you choose an Abstract Class?
Answer: When classes share common code and state.
14. Can an abstract class have constructors?
Answer: Yes.
15. Can an interface have fields?
Answer: No.

Tricky Questions

66. Can we override a static method?
Answer: No.
67. Can a constructor be virtual?
Answer: No.
68. Can a sealed class be inherited?
Answer: No.
69. Can a static class have a constructor?
Answer: Yes, but only a static constructor.
70. Can an interface be instantiated?
Answer: No.
71. Why is Object the base class in C#?
Answer: Every .NET type ultimately derives from System.Object.
72. Difference between throw and throw ex?
Answer: throw preserves the original stack trace; throw ex resets it.
73. Why use using?
Answer: It automatically calls Dispose() to release resources.
74. What is Reflection?
Answer: Reflection inspects and invokes types, members, and assemblies at runtime.
75. What is an Extension Method?
Answer: It adds methods to existing types without modifying their source code.

C# Deep Questions

16. Why is string immutable?
Answer: For thread safety, security, and performance through string interning.
17. Why is String a reference type?
Answer: Because it stores objects on the heap, even though its value cannot be modified.
18. Difference between == and Equals()?
Answer: == compares values for strings, while Equals() checks logical equality.
19. Why use StringBuilder?
Answer: It avoids creating multiple string objects during modifications.
20. What is Boxing?
Answer: Boxing converts a value type into an object (reference type).
21. What is Unboxing?
Answer: Unboxing converts an object back into its original value type.
22. Why is Boxing expensive?
Answer: Because it allocates memory on the heap.
23. Difference between ref, out, and in?
Answer: ref requires initialization, out returns a value, and in passes a read-only reference.
24. Difference between const and readonly?
Answer: const is compile-time, while readonly is runtime.
25. Difference between var, dynamic, and object?
Answer: var is compile-time inferred, dynamic is runtime resolved, and object is the base type.

Memory

26. Stack vs Heap?
Answer: Stack stores local variables; Heap stores objects.
27. Who manages Heap memory?
Answer: The Garbage Collector.
28. What is Garbage Collection?
Answer: It automatically frees memory occupied by unreachable objects.
29. What causes a Memory Leak?
Answer: Objects remain referenced and cannot be collected by the Garbage Collector.
30. What is Dispose()?
Answer: Dispose releases unmanaged resources immediately.
31. Why implement IDisposable?
Answer: To ensure timely cleanup of unmanaged resources.

Collections

32. Array vs List?
Answer: Arrays are fixed-size; Lists are dynamically resizable.
33. List vs LinkedList?
Answer: List is faster for indexing; LinkedList is faster for insertions and deletions.
34. Dictionary vs Hashtable?
Answer: Dictionary is generic and type-safe; Hashtable is non-generic.
35. HashSet?
Answer: HashSet stores only unique values.
36. Queue?
Answer: Queue follows FIFO.
37. Stack?
Answer: Stack follows LIFO.

LINQ

38. IEnumerable vs IQueryable?
Answer: IEnumerable executes in memory; IQueryable executes on the data source.
39. Deferred Execution?
Answer: LINQ queries execute only when enumerated.
40. Immediate Execution?
Answer: Methods like ToList() execute the query immediately.
41. First vs FirstOrDefault?
Answer: First() throws if empty; FirstOrDefault() returns the default value.
42. Single vs First?
Answer: Single() expects exactly one result; First() returns the first match.

Async Programming

43. async/await?
Answer: They enable asynchronous, non-blocking code.
44. Task vs Thread?
Answer: Task is a higher-level abstraction that may use one or more threads.
45. Task vs ValueTask?
Answer: ValueTask reduces allocations when operations often complete synchronously.

Dependency Injection

46. What is Dependency Injection?
Answer: DI provides dependencies from outside instead of creating them inside the class.
47. Why use DI?
Answer: It reduces coupling and improves testability.
48. Singleton Lifetime?
Answer: One instance for the entire application.
49. Scoped Lifetime?
Answer: One instance per HTTP request.
50. Transient Lifetime?
Answer: A new instance every time it is requested.

SOLID

51. SRP?
Answer: A class should have only one reason to change.
52. OCP?
Answer: Software should be open for extension but closed for modification.
53. LSP?
Answer: A derived class must be replaceable with its base class.
54. ISP?
Answer: Clients should not depend on methods they don't use.
55. DIP?
Answer: Depend on abstractions instead of concrete implementations.

Design Patterns

56. Singleton?
Answer: Ensures only one instance of a class exists.
57. Factory?
Answer: Encapsulates object creation logic.
58. Repository?
Answer: Separates data access from business logic.
59. Unit of Work?
Answer: Groups multiple database operations into one transaction.
60. Strategy?
Answer: Encapsulates interchangeable algorithms behind a common interface.
61. Builder?
Answer: Constructs complex objects step by step.
62. Adapter?
Answer: Makes incompatible interfaces work together.
63. Observer?
Answer: Notifies dependent objects when the subject changes.
64. Mediator?
Answer: Centralizes communication between objects.
65. Decorator?
Answer: Adds behavior to an object without modifying its class.

Angular Interview Questions (5+ Years)

1. What is Angular?

Answer: Angular is a TypeScript-based front-end framework for building single-page applications.

2. What is SPA?

Answer: A Single Page Application loads one HTML page and updates content dynamically.

3. Component?

Answer: A component is the basic building block of an Angular application.

4. Module?

Answer: A module groups related components, directives, pipes, and services.

5. Service?

Answer: A service contains reusable business logic shared across components.

6. Dependency Injection?

Answer: DI injects services into components without creating them manually.

7. Data Binding?

Answer: Data binding synchronizes data between the component and the view.

8. Types of Data Binding?

Answer: Interpolation, Property, Event, and Two-way binding.

9. Interpolation?

Answer: Displays component data using {{ }}.

10. Property Binding?

Answer: Binds a component value to an HTML property.

11. Event Binding?

Answer: Calls a method when an event occurs.

12. Two-way Binding?

Answer: Synchronizes data between the UI and the component using [(ngModel)].

13. Directive?

Answer: Directives change the behavior or appearance of DOM elements.

14. Structural Directive?

Answer: Changes the DOM layout (e.g., *ngIf, *ngFor).

15. Attribute Directive?

Answer: Changes the appearance or behavior of an existing element.

16. Pipe?

Answer: Pipes transform data for display.

17. Pure Pipe?

Answer: Executes only when the input reference changes.

18. Observable?

Answer: An Observable emits asynchronous data streams over time.

19. Subject?

Answer: A Subject is both an Observable and an Observer.

20. BehaviorSubject?

Answer: It stores the latest value and emits it to new subscribers.

21. Observable vs Promise?

Answer: Observable supports multiple values and cancellation; Promise returns a single value.

22. Lazy Loading?

Answer: Loads feature modules only when they are needed.

23. Route Guard?

Answer: Controls access to routes before navigation.

24. CanActivate?

Answer: Determines whether a route can be activated.

25. Interceptor?

Answer: Intercepts HTTP requests and responses globally.

26. Change Detection?

Answer: Angular updates the view when component data changes.

27. OnPush Strategy?

Answer: Reduces unnecessary change detection for better performance.

28. ViewChild?

Answer: Accesses child components or DOM elements.

29. ngOnInit?

Answer: Called once after component initialization.

30. ngOnDestroy?

Answer: Used to clean up subscriptions and resources.

31. Why unsubscribe?

Answer: To prevent memory leaks.

32. TrackBy?

Answer: Improves *ngFor performance by tracking items using a unique key.

33. Reactive Forms?

Answer: Forms managed entirely in TypeScript.

34. Template-driven Forms?

Answer: Forms managed primarily in the HTML template.

35. Reactive vs Template Forms?

Answer: Reactive forms are more scalable and suitable for complex forms.

SQL Server Interview Questions

1. Primary Key?

Answer: Uniquely identifies each row and does not allow NULL values.

2. Foreign Key?

Answer: Maintains relationships between tables.

3. Clustered Index?

Answer: Determines the physical order of data.

4. Non-Clustered Index?

Answer: Stores index data separately from the table.

5. Clustered vs Non-Clustered?

Answer: Clustered changes data order; Non-clustered does not.

6. Index?

Answer: Improves query performance.

7. Composite Index?

Answer: An index on multiple columns.

8. Stored Procedure?

Answer: A precompiled SQL program stored in the database.

9. View?

Answer: A virtual table based on a SQL query.

10. CTE?

Answer: A temporary result set used within a single query.

11. Temp Table?

Answer: A temporary table stored in tempdb.

12. Table Variable?

Answer: A lightweight temporary table stored in memory or tempdb.

13. DELETE vs TRUNCATE?

Answer: DELETE logs each row and can use WHERE; TRUNCATE removes all rows quickly.

14. TRUNCATE vs DROP?

Answer: TRUNCATE removes data; DROP removes the table.

15. INNER JOIN?

Answer: Returns matching rows from both tables.

16. LEFT JOIN?

Answer: Returns all rows from the left table and matching rows from the right.

17. RIGHT JOIN?

Answer: Returns all rows from the right table and matching rows from the left.

18. FULL JOIN?

Answer: Returns all matching and non-matching rows.

19. UNION vs UNION ALL?

Answer: UNION removes duplicates; UNION ALL keeps duplicates.

20. ROW_NUMBER()?

Answer: Assigns a unique sequential number to each row.

21. RANK()?

Answer: Assigns ranks with gaps for duplicate values.

22. DENSE_RANK()?

Answer: Assigns ranks without gaps.

23. Transactions?

Answer: A transaction groups SQL operations into one logical unit.

24. ACID?

Answer: Atomicity, Consistency, Isolation, Durability.

25. Deadlock?

Answer: Two transactions wait indefinitely for each other's resources.

26. Normalization?

Answer: Organizes data to reduce redundancy.

27. Denormalization?

Answer: Adds redundancy to improve read performance.

28. HAVING vs WHERE?

Answer: WHERE filters rows; HAVING filters grouped results.

29. EXISTS vs IN?

Answer: EXISTS stops after finding a match; IN compares against a list.

30. Execution Plan?

Answer: Shows how SQL Server executes a query.

Azure Interview Questions

1. What is Azure?

Answer: Azure is Microsoft's cloud computing platform.

2. Azure App Service?

Answer: Hosts web applications without managing servers.

3. Azure Function?

Answer: A serverless service that runs code on demand.

4. Azure Storage Account?

Answer: Provides cloud storage services.

5. Blob Storage?

Answer: Stores files such as images, videos, and documents.

6. Queue Storage?

Answer: Stores messages for asynchronous processing.

7. Azure Service Bus?

Answer: A reliable messaging service for enterprise applications.

8. Azure SQL Database?

Answer: A fully managed cloud SQL database.

9. Azure Key Vault?

Answer: Securely stores secrets, certificates, and encryption keys.

10. Azure App Configuration?

Answer: Centralizes application configuration settings.

11. Azure API Management?

Answer: Publishes, secures, and monitors APIs.

12. Azure Application Insights?

Answer: Monitors application performance and failures.

13. Azure Monitor?

Answer: Collects and analyzes metrics and logs.

14. Azure Virtual Machine?

Answer: A cloud-based virtual server.

15. Azure DevOps?

Answer: Provides CI/CD pipelines, repositories, and project management.

16. CI/CD?

Answer: Automates code integration, testing, and deployment.

17. Azure Load Balancer?

Answer: Distributes network traffic across multiple servers.

18. Azure Front Door?

Answer: Provides global load balancing and web application acceleration.

19. Azure CDN?

Answer: Delivers content from locations closest to users.

20. Managed Identity?

Answer: Allows Azure services to authenticate securely without storing credentials.

21. Authentication vs Authorization?

Answer: Authentication verifies identity; Authorization determines permissions.

22. Azure AD (Microsoft Entra ID)?

Answer: Provides identity and access management for applications and users.

23. Azure Redis Cache?

Answer: Improves application performance by caching frequently accessed data.

24. Azure Logic Apps?

Answer: Automates workflows by connecting services and applications.

25. Why use Azure over on-premises?

Answer: Azure provides scalability, high availability, managed services, and pay-as-you-go pricing.